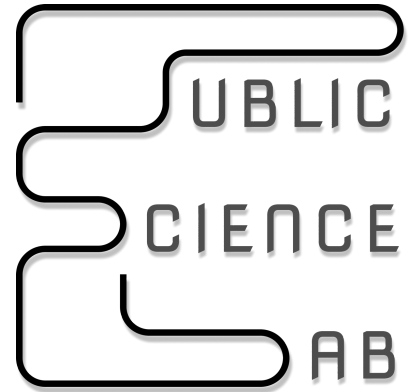




European Research Council
Established by the European Commission



Humans in Ontologies

Sabina Leonelli



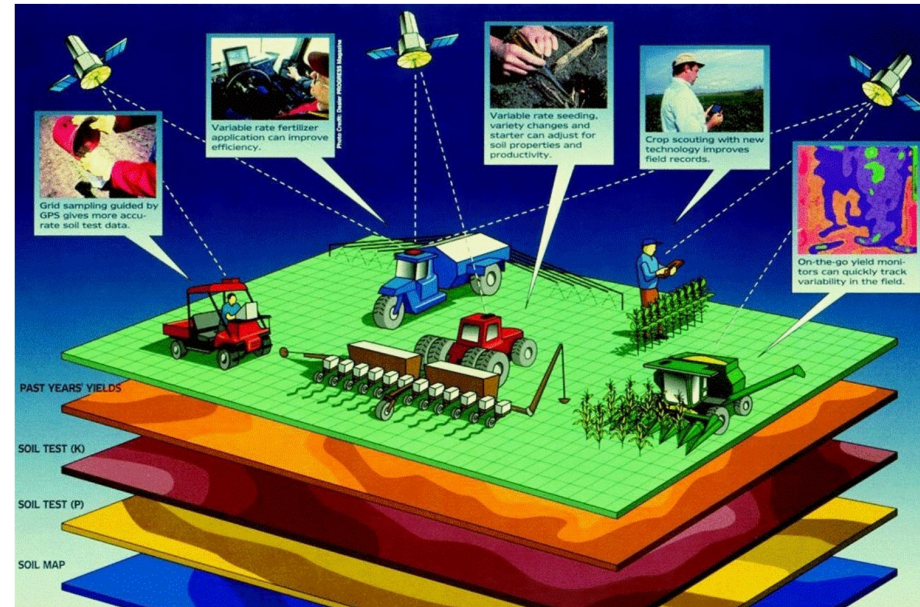
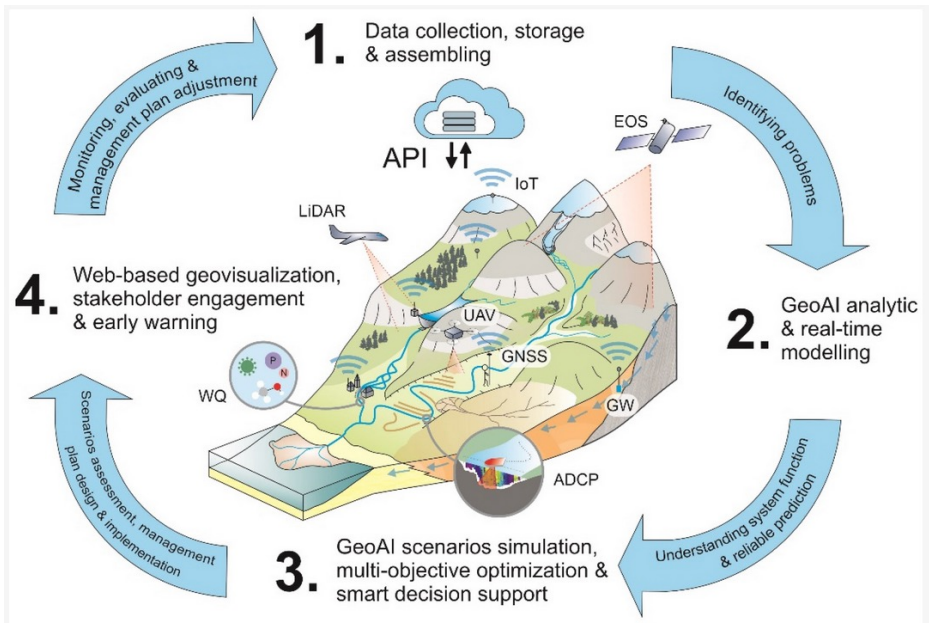
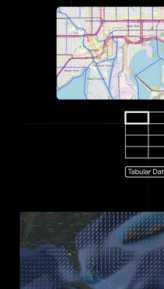


Figure 1: GIS in Agriculture: Map layers such as soil chemistry, soil type and topography (and derived maps such as soil drainage), previous yields, etc. can provide important information for field management

<https://gis-university.com/gis-in-agriculture>

Geospatial Reasoning: Unlocking insights with generative AI and multiple foundation models









**RESEARCH
PROGRAM ON
Roots, Tubers
and Bananas**



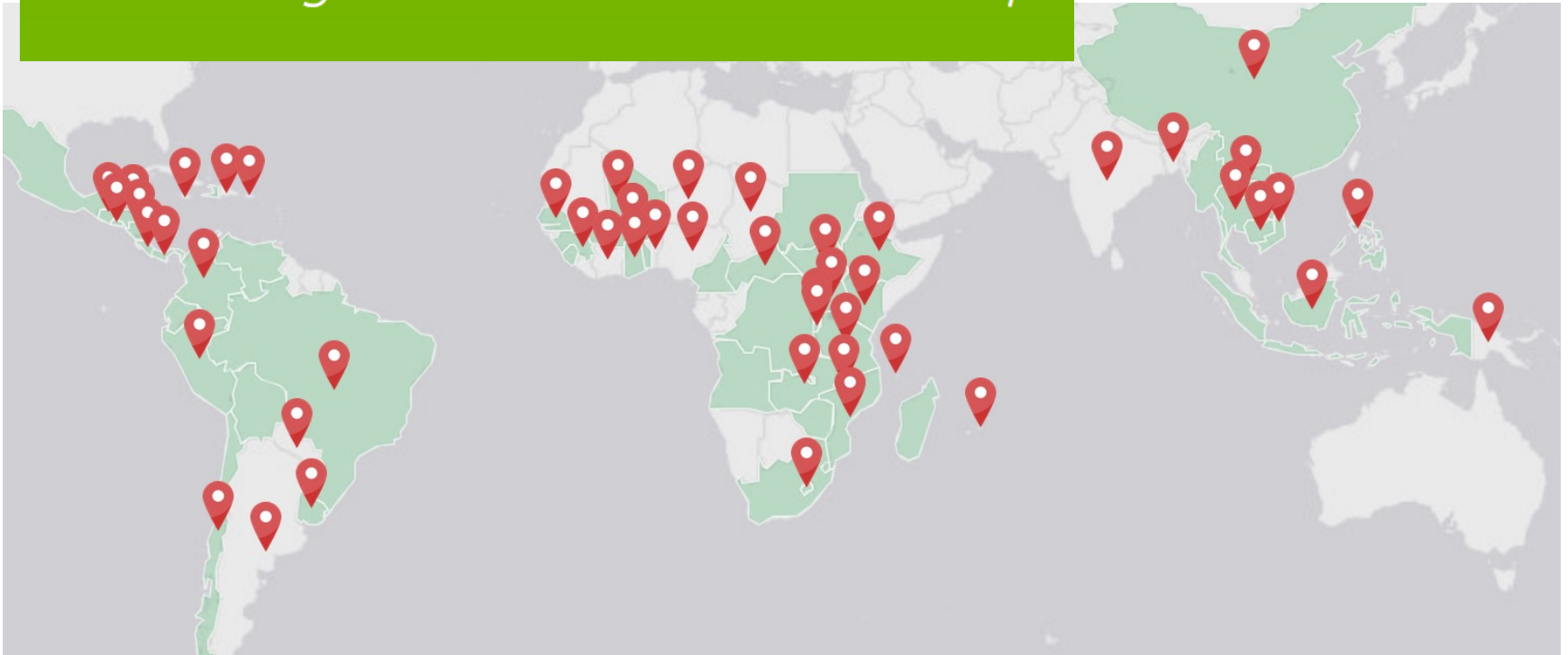
CGIAR Centres

Formerly: Consultative Group for International Agricultural Research.

>10.000 staff, 96 countries



A Global Agricultural Research Partnership



The challenge of descriptors for crop traits

- Identifying “standard” and “universal” descriptors is marred by complexity of
 - biology (diversity and variability of traits to be described)
 - technology (diversity of methods and tools to measure traits)
 - human interactions with plants across the globe (as food, culture, commodities)
- Complexity is conceptualised and presented as a “problem”, to which standard descriptors may provide a “solution” (for details on the history of CGIAR descriptors: Curry and Leonelli 2024)
- Yet descriptors themselves need to take account of diversity in rationales and approaches used to capture the morphology of plants, esp. beyond academic research



Table 2. Descriptors categories and their classes used in the morphological characterization of cassava germplasm, Chapadinha, MA, 2013.

S/N	Plant descriptors	Given categories
1	Branching habit	1-Erect; 2-Dichotomous; 3-Trichotomous and 4-Tetrachotomous.
2	Type of plant	1-Open; 2-Umbrella type and 3-Compact
	Leaf descriptors	
3	Apical leaf color	1- Light green; 2- Dark green; 3- Purplish-green and 4- Purple.
4	Pubescence of apical bud	1-Present and 2- Absente.
5	Petiole color	1-Yellowish-green; 2- Green; 3-Redish-green; 4- greenish-red; 5-red and 6-Purple
6	Developed leaf color	1-Light-green; 2-Dark-green; 3-Purplish-green and 4-purple
7	Terminal branches color	1-Light-green; 2-Dark-green; 3-Purplish-green and 4-purple
8	Leave's rib color	1- Green; 2-Redish-green; and 3- Greenish-red
9	petiole position	1-Tilted up; 2-Horizontal; 3-Angled down and 4-Irregular
10	Prominence of leaf scars	1-Without prominence and 2-Proeminent.
	Stem Descriptors	
11	Color of stem cortex	1-Light yellow; 2-Light green; 3-Green and 4-Dark green.
12	Length of phyllotaxis	1-Short; 2-Middle and 3-Large.
13	External Color of steam	1-Orange; 2-Yellowish-green; 3-Golden; 4-Light brown; 5-Gray; 6- Silvery; 7- Gray; 8- Silvery; 9- Dark brown.
14	Color of stem epidermis	1- Cream; 2- Light brown; 3- Dark brown; 4- Yellow.
15	Growth habit of the stem	1-Straight and 2-Forked.
	Root descriptors	
16	Presence of peduncle in roots	1-Present and 2-Absent.
17	External color of roots	1-White; 2-Yellow; 3-Light brown; 4-Brown and 5-Dark brown.
18	Color of root Cortex	1-White; 2-Yellow and 3-Pinkish.
19	Texture of root epidermis	1-Smooth and 2-Rough.
20	Constriction of roots	1-Absent; 2-Little or none and 3-Average.
21	Root shape	1-Conical; 2-Cylinder and 3-Spindle.
22	Highlight pellicle from roots	1-Easy release and 2- Difficult release.
23	Highlight of roots cortex	1-Easy release and 2-Difficult release.
24	Position of roots	1-Horizontal and 2- Vertical tendency.

Challenges from the field/1

- **Variability within and across crop types**
 - What counts as an individual, a variety, an accession, an ecotype, a strain (especially for clonal reproduction)?
 - Soil, environment, nurture all strongly affect phenotype: how to pick out representative/‘normal’ traits?
 - Which individuals are representative of a plot?
- **Naming of varieties**
 - Locally named from person who gives the clone, so breeders need to reconstruct clone origin by tracking back the gifting of clones
 - Genetic markers used to determine ‘real’ name – however, unclear taxonomy of phenotype, and big implications for local culture/markets
 - Uneasy relationship to traditional taxonomy

Challenges from the field/2

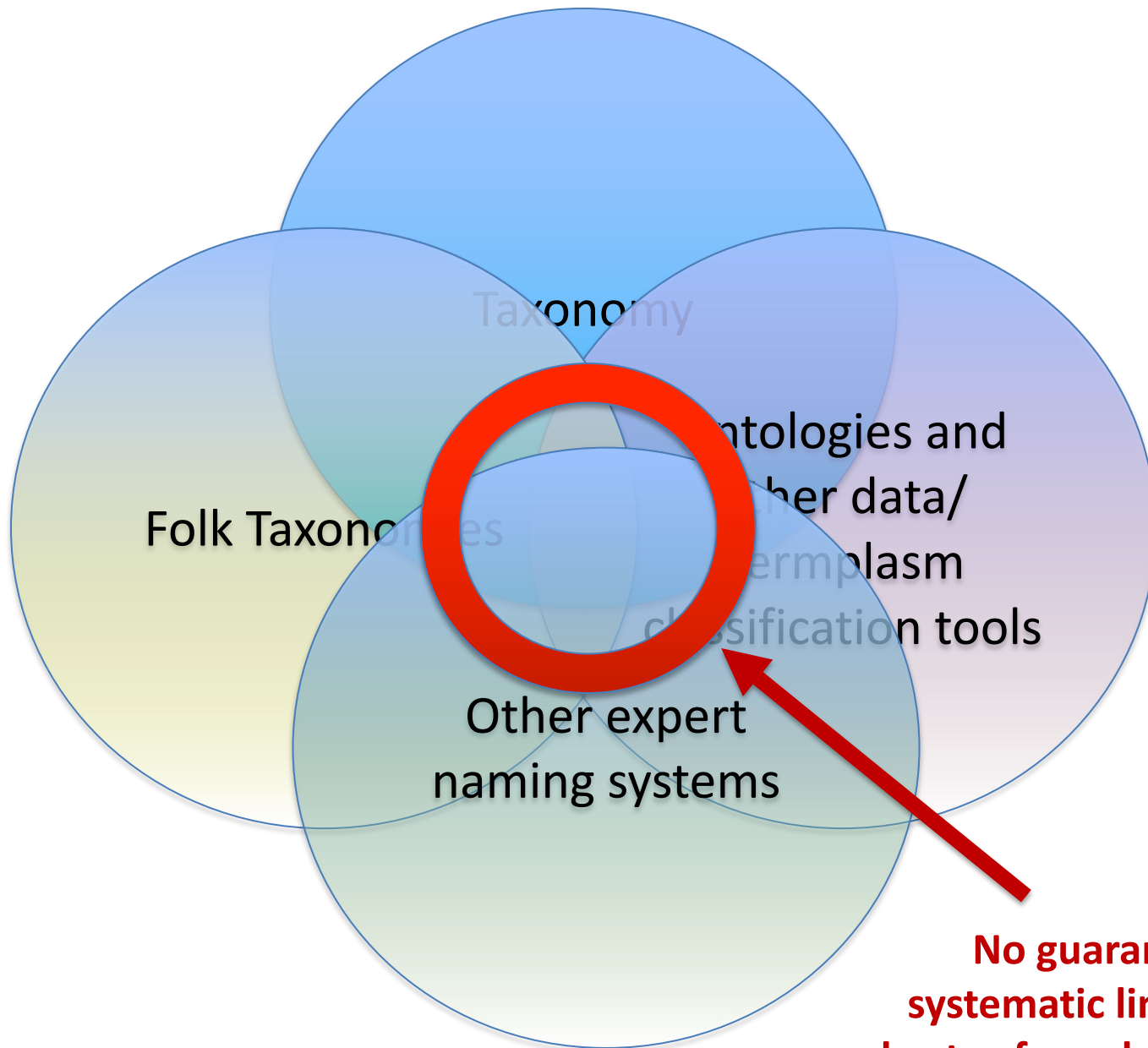
- **Variability in skills and measuring methods for traits data collection**
 - how plot is measured and sampled; color assessment; how harvest is collected (the 'standard cut'); counting flowers
- **Shifting temporalities of data collection**
 - Time of flowering and harvest can vary considerably within and across trials
 - Constant monitoring and no standardization
 - Processing and management of germplasm materials



Challenges from the field/3

- **Identification of valuable traits: which trait matters and for whom?**
 - Trialing varieties can take 6-7 years, crucial to consult with farmers regularly
 - Gender differences & complex local mediations around choice of variety for any given trial
- = **Challenge of instituting global standards for local, situated procedures**





**No guarantee of
systematic links among
subsets of overlapping names!**

Plant Naming and the Crop Ontology

The Crop Ontology

- Standardizing and enriching the representation of crop traits and trait-attributes across crop species and databases
- Started with crop-specific traits for chickpea (*Cicer arietinum*), maize (*Zea mays*), potato (*Solanum tuberosum*), rice (*Oryza sativa*), sorghum (*Sorghum* spp.) and wheat (*Triticum* spp.). In 2010, publication of plant-structure and anatomy-related terms for banana (*Musa* spp.) and rice; start of expansion to cassava, yam, etc.
- Multi-crop passport terms are included as controlled vocabularies for sharing information on germplasm

Stem Color
CO_334:0000062

Root Shape
CO_334:0000020

Root Neck Length
CO_334:0000022

Outer Skin Color
CO_334:0000064

Root Number
CO_334:0000011



Crop Ontology on cassava: 365 traits in 2021 (from 120 in 2011)

Ontology browser

Search and browse ontologies

Find exact ID |

Search for text CO (Cassava Trait Ontology)

CO:0000000 CGIAR cassava trait ontology

is_a CO:0000001 Agronomic trait

is_a CO:0000002 Morphological trait

is_a CO:0000003 Physiological trait

is_a CO:0000004 Quality trait

is_a CO:0000005 Stress trait

is_a CO:0000006 abiotic stress

is_a CO:0000007 biotic stress

is_a CO:0000027 bacterial disease

is_a CO:0000029 fungal disease

is_a CO:0000038 Cassava

is_a CO:0000032 Cassava

is_a CO:0000030 insect damage

is_a CO:0000028 viral disease

GO:0008150 biological_process

GO:0003674 molecular_function

GO:0005575 cellular_component

PO:0025131 plant anatomical entity

PO:0009012 plant growth and development

SO:0000400 sequence_attribute

SO:0001060 sequence_variant

SO:0000110 sequence_feature

CO:0000032 'Cassava anthracnose disease severity'

Cvterm details

Term id 0000032

Term name Cassava anthracnose disease severity

Term definition Severity of the Cassava anthracnose disease caused by *Colletotrichum gloeosporioides* f. sp.

Comment Symptoms: Cankers on the stems and bases of leaf petioles.

Synonyms

"Cassava anthracnose disease severity - Cassavabase" EXACT []

"CADSev" EXACT []

CADS EXACT

"CADS" EXACT []

Definition dbxrefs

CO:curators

Definition dbxrefs

TO:0000439

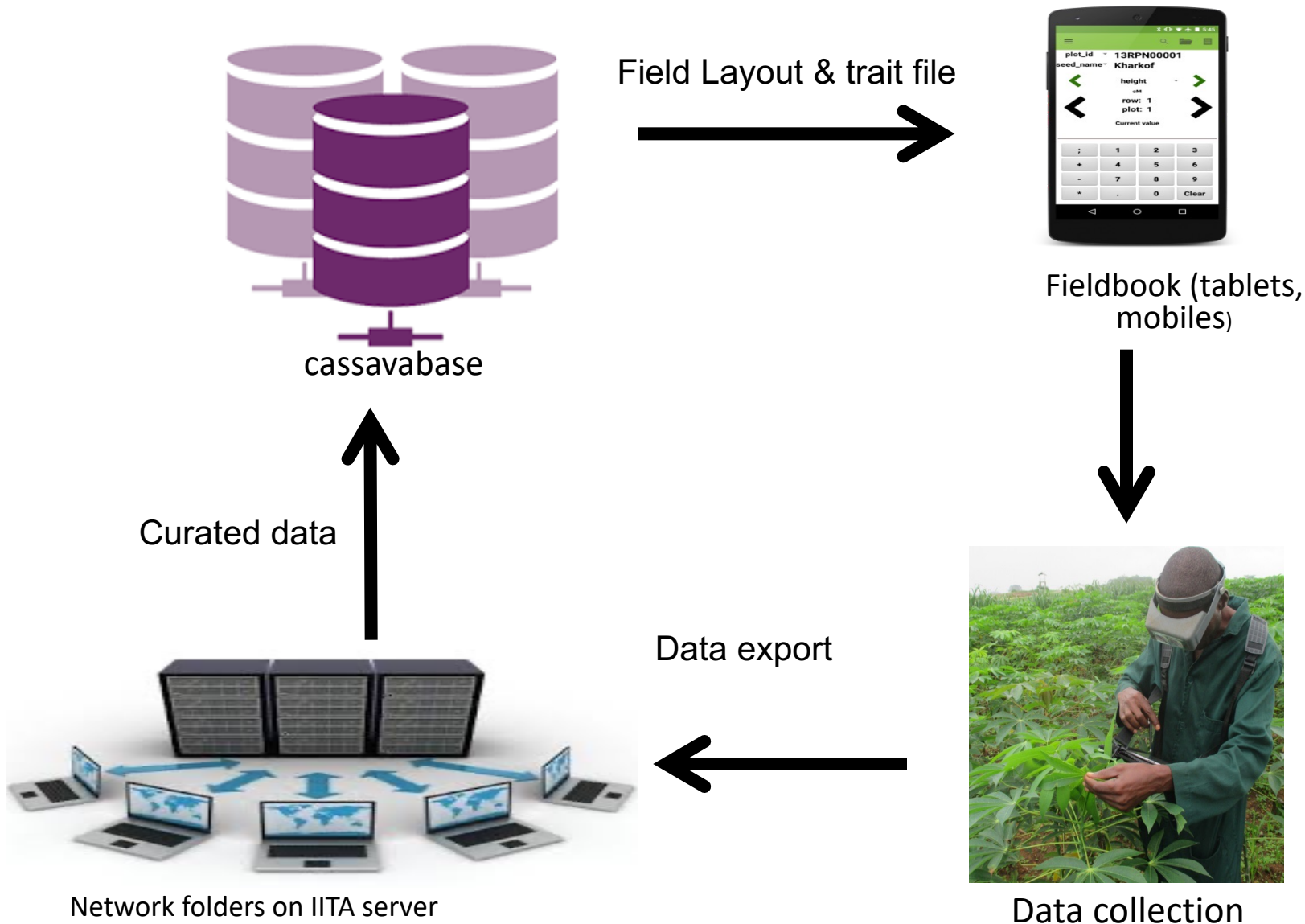
Classification of cassava ontology terms

The ontology terms for cassava are grouped into the following classes

- Agronomic trait
- Morphological trait
- Physiological trait
- Stress trait
- Quality trait

How Does the Crop Ontology Complement Descriptors?

[1] Digitalisation of Phenotype Workflow



Data collection using fieldbook



10:48

plot_id 2017-16.GS.CO.C1.AYT.58.IB-rep1-IITA-TM
accession IITA-TMS-IBA9821

< SPRTCT6 |CO:0.. >
sprout count at six-month

plot: 1004
rep: 1

; 1 2 3
+ 4 5 6
- 7 8 9
* . 0

!

11:17

< ✓ 123 SPROUT |CO:0000008
✓ 123 SPRTCT1 |CO:0000213
✓ 123 SPRTCT3 |CO:0000214
✓ 123 SPRTCT6 |CO:0000215
✓ VIGOR |CO:0000009
✓ 123 CMD1S |CO:0000191
✓ 123 CMD3S |CO:0000192
✓ 123 CMD6S |CO:0000194
✓ 123 CMD1I |CO:0000195
✓ 123 CMD3I |CO:0000196
✓ 123 CMD6I |CO:0000198
✓ 123 CBB3S |CO:0000175

+

10:52

plot_id 2017-16.GS.CO.C1.AYT.58.IB-rep1-IITA-TM
accession IITA-TMS-IBA9821

< CBB3S |CO:000. >
cassava bacterial blight severity 3-month evaluation
Manihotis symptoms rated from 1 = (clean, no
infection) to 5 = (extremely severe, severely diseased)
evaluated between two and three months after
planting.

plot: 1004
rep: 1

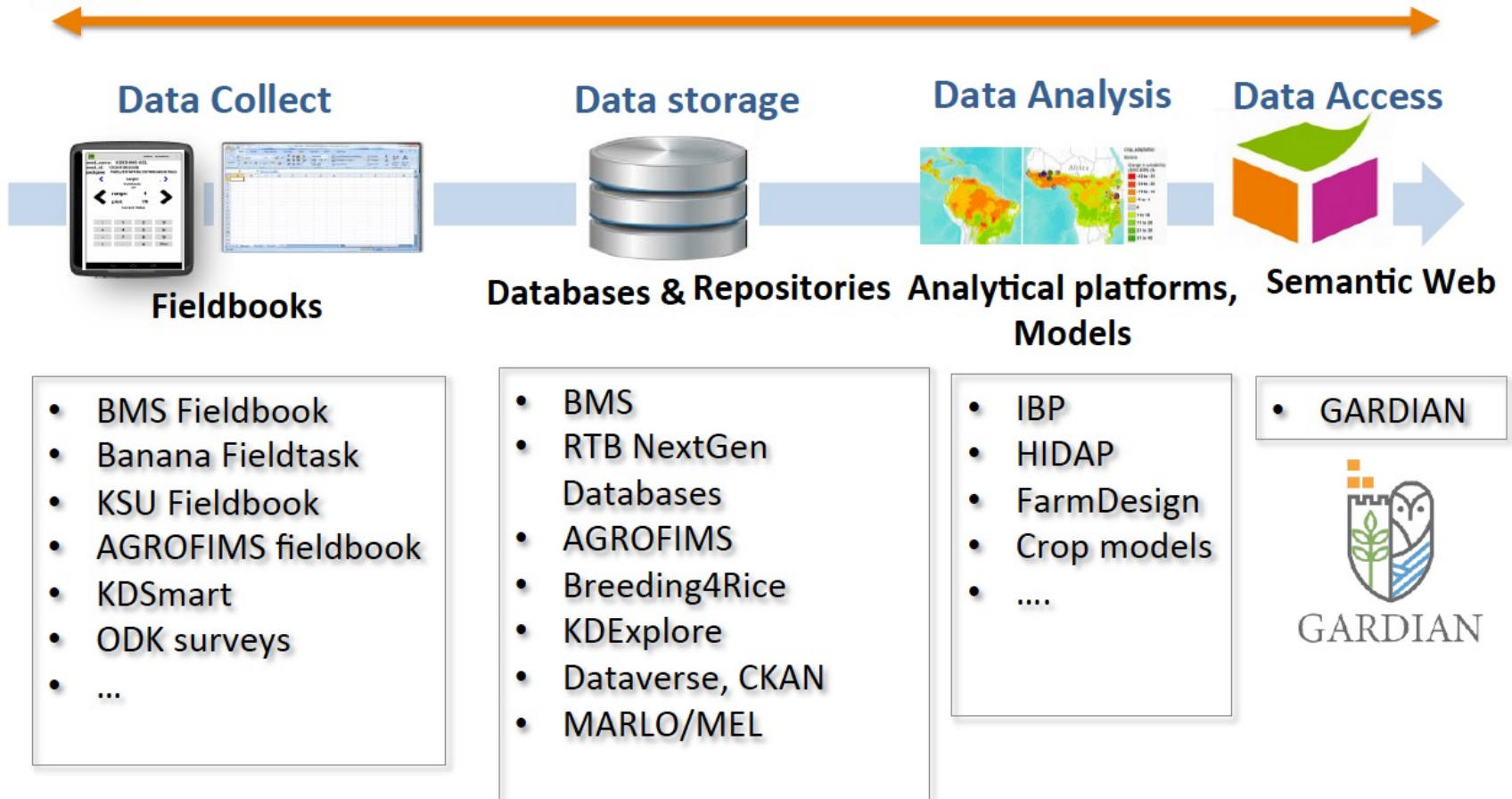
; 1 2 3
+ 4 5 6
- 7 8 9

!

Phenotypic Data Harmonization along the data management workflow



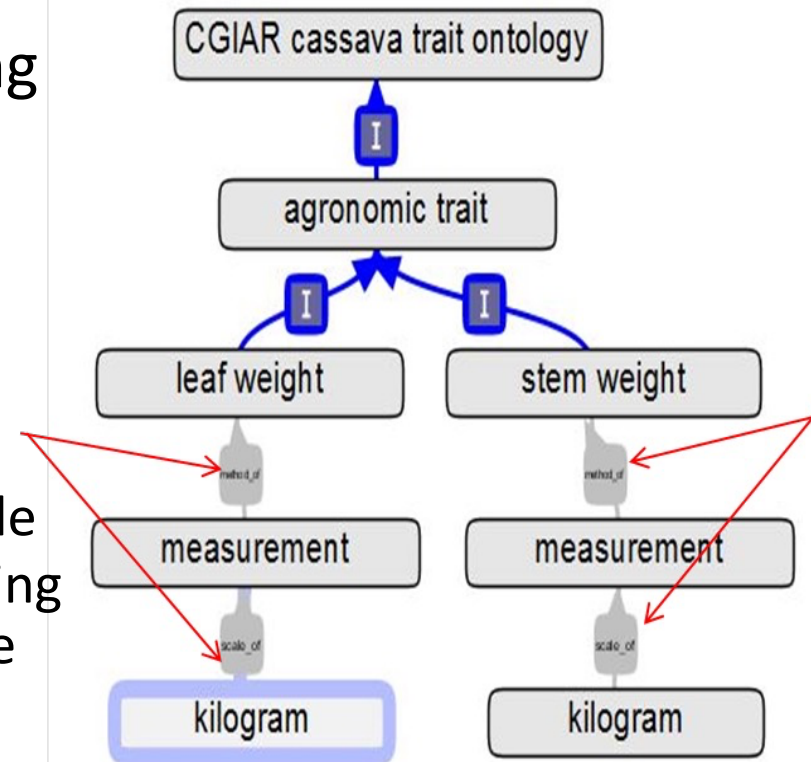
Semantics for harmonization & interoperability



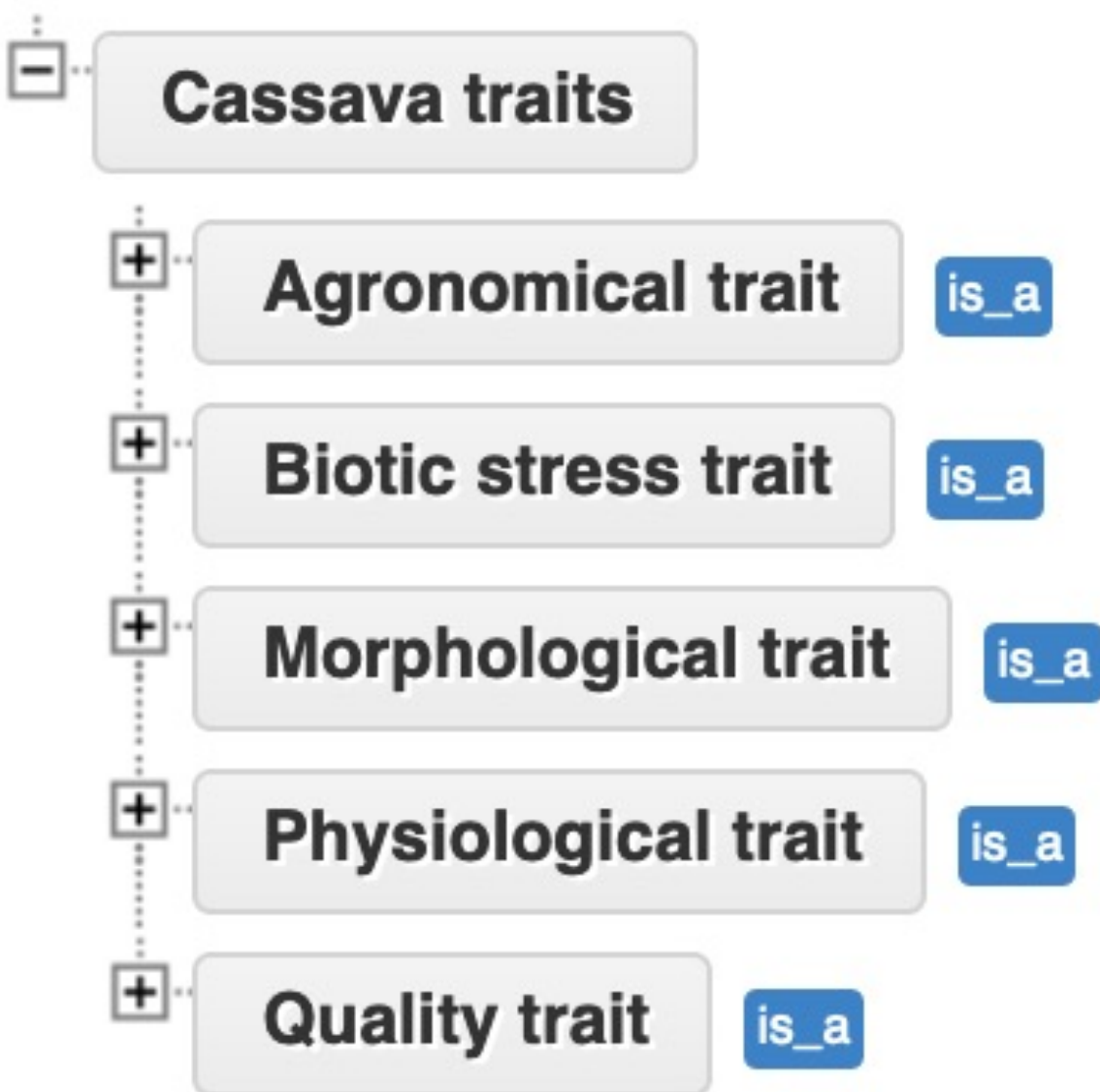
[2] Linking naming with method and skills

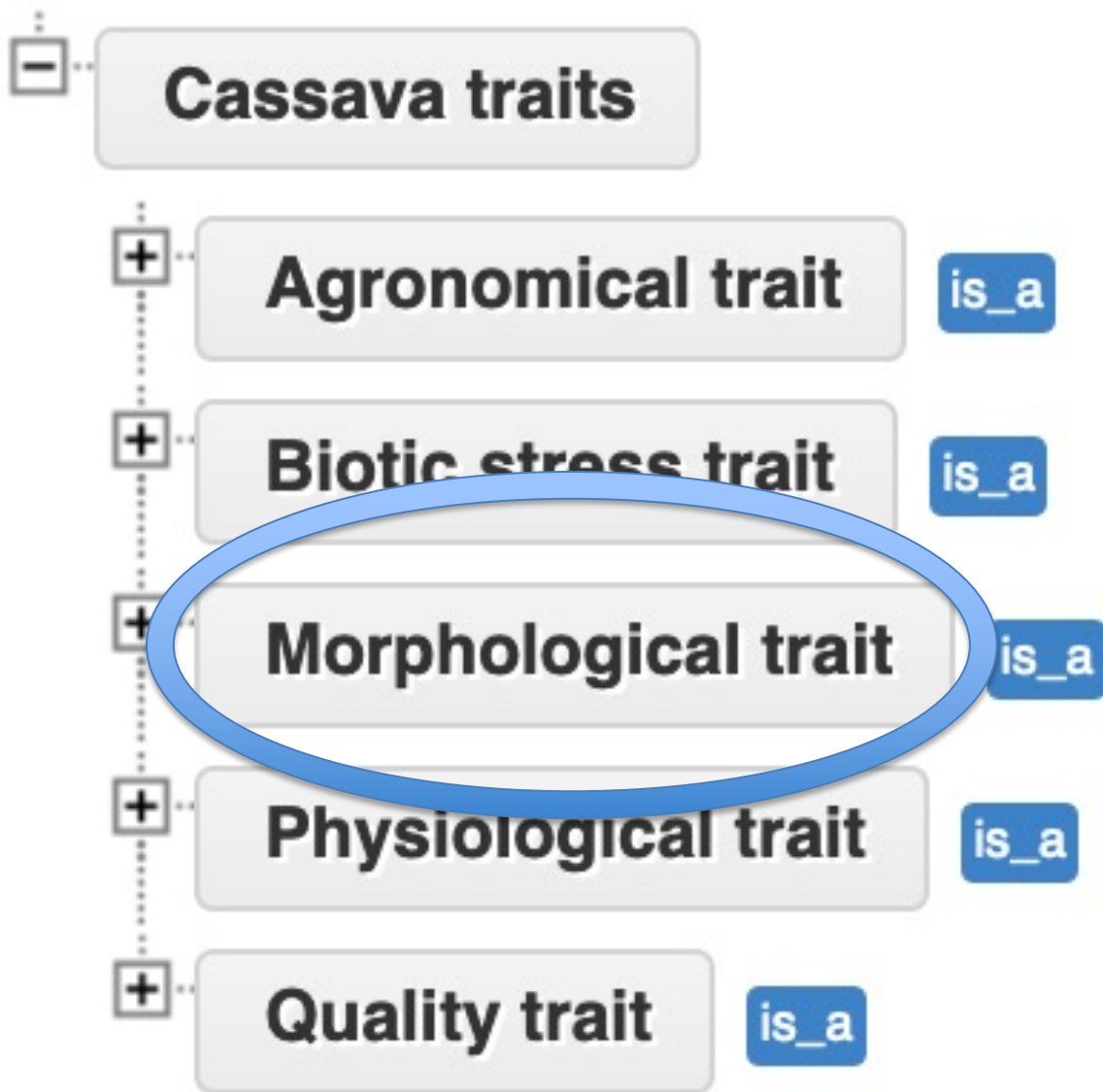
“one of the reasons cassava databases are working is because of the ontology work, the attempt to have a standardized system for collecting data” (informer 32)
But what kind of standardization effort is this?

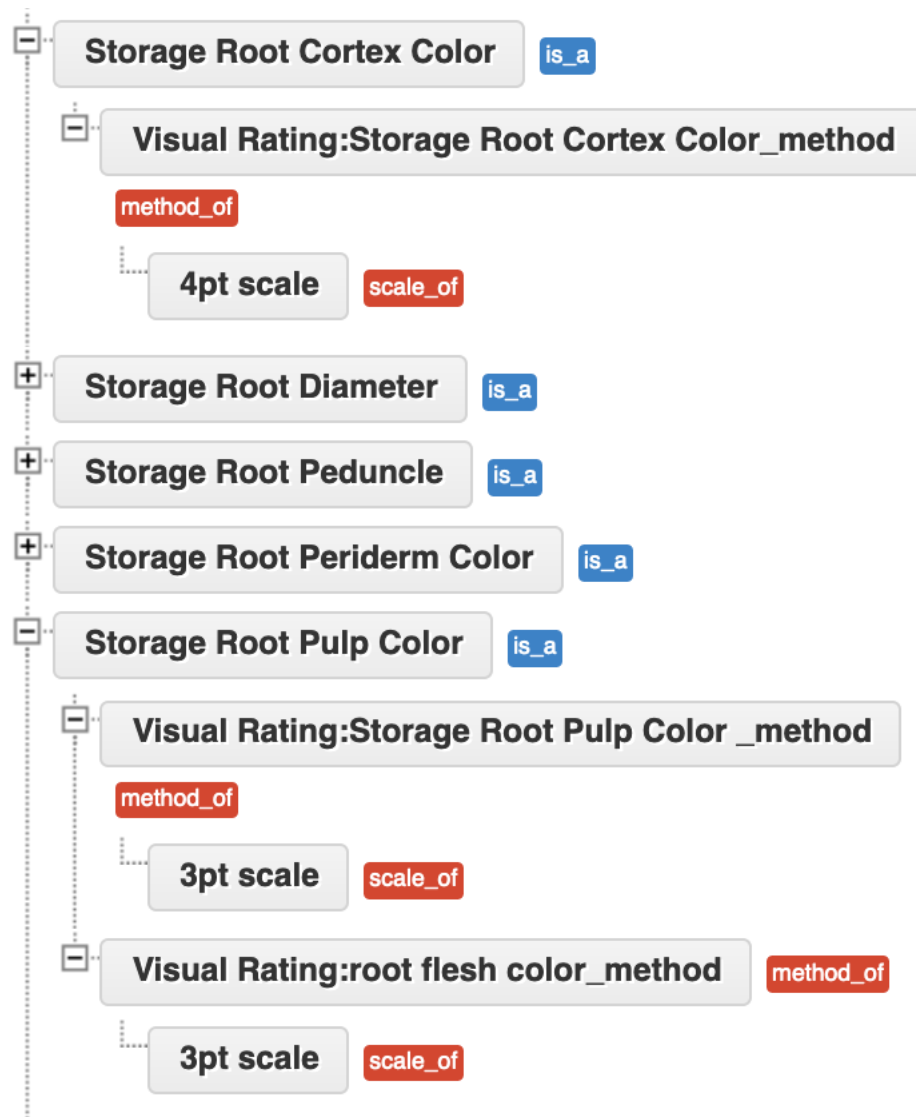
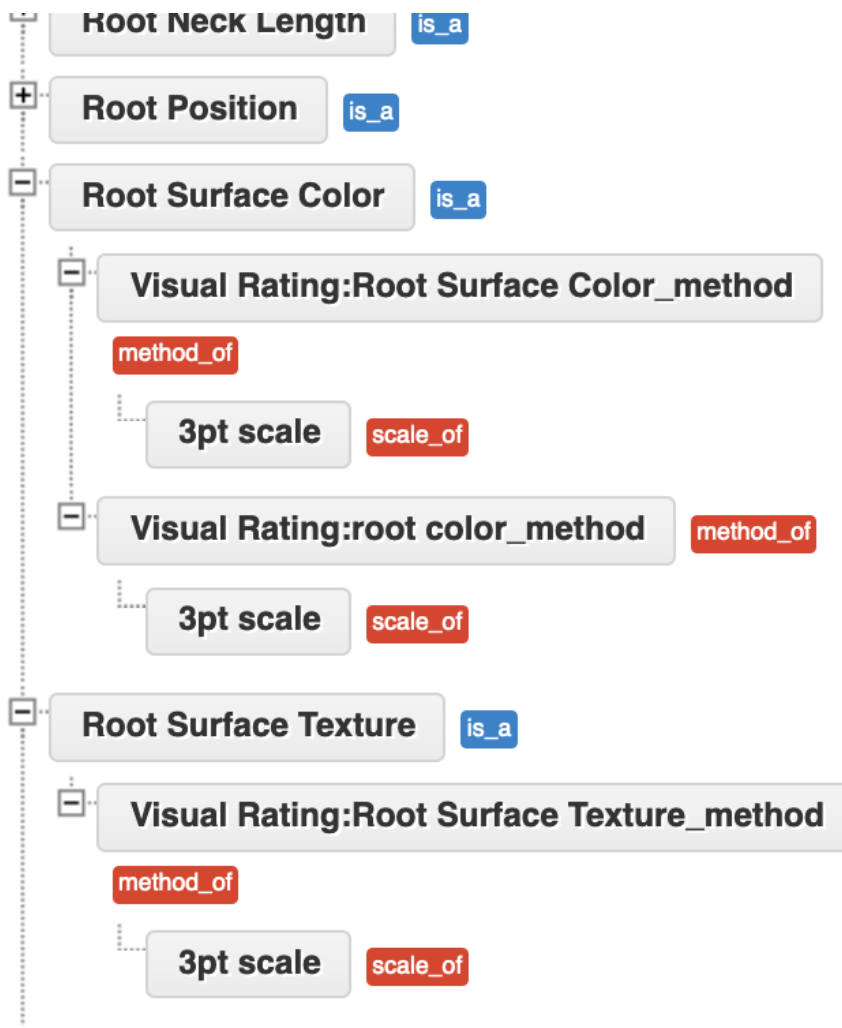
- Main relation among traits is **‘variable of’** (hard in OBI format)
- Traits are classified by differentiating between Trait Definition, Measurement Method and Scale (‘scale_of’, ‘measured_by’)
- **Ability to deal with variation in approaches and skills applied to estimates and measures.** E.g. multiple shades of colors: captured through changing skill (so color is still yellow overall, but one can use skill of discerning color)

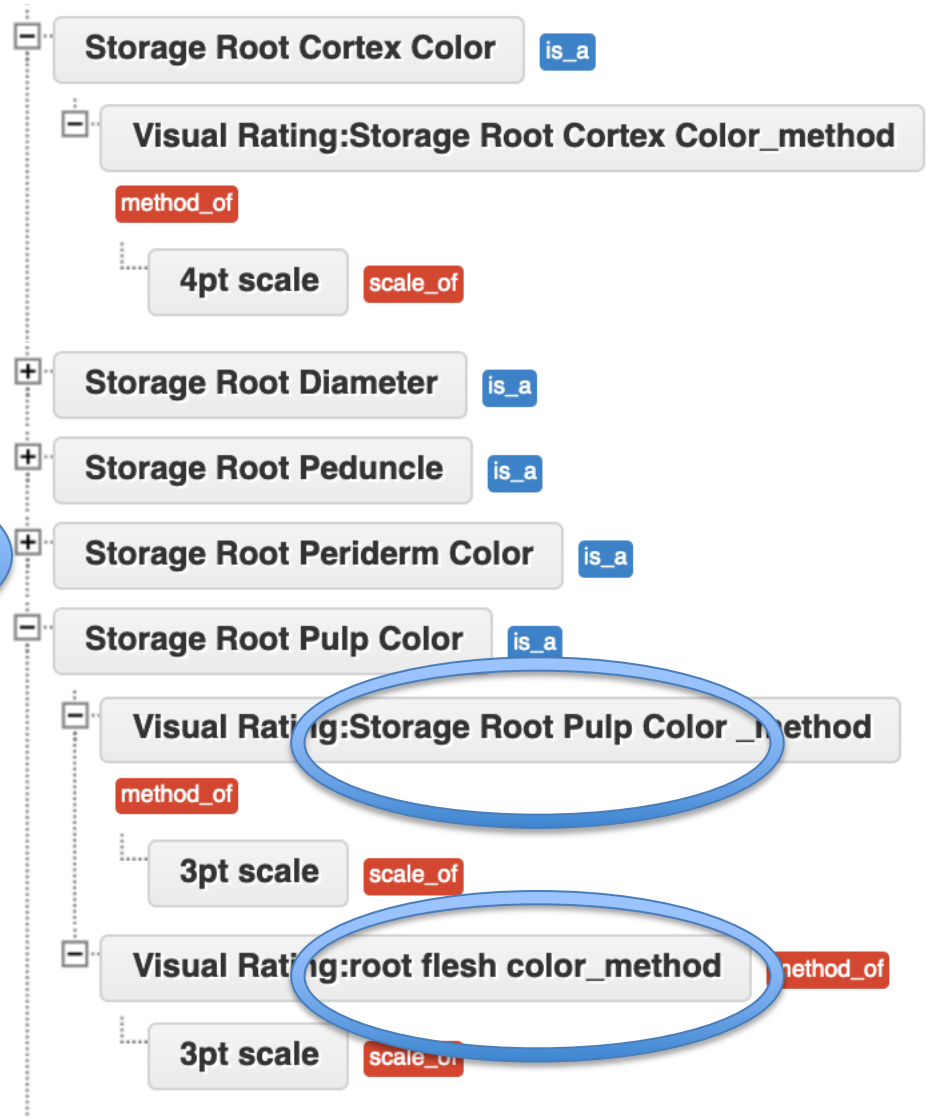
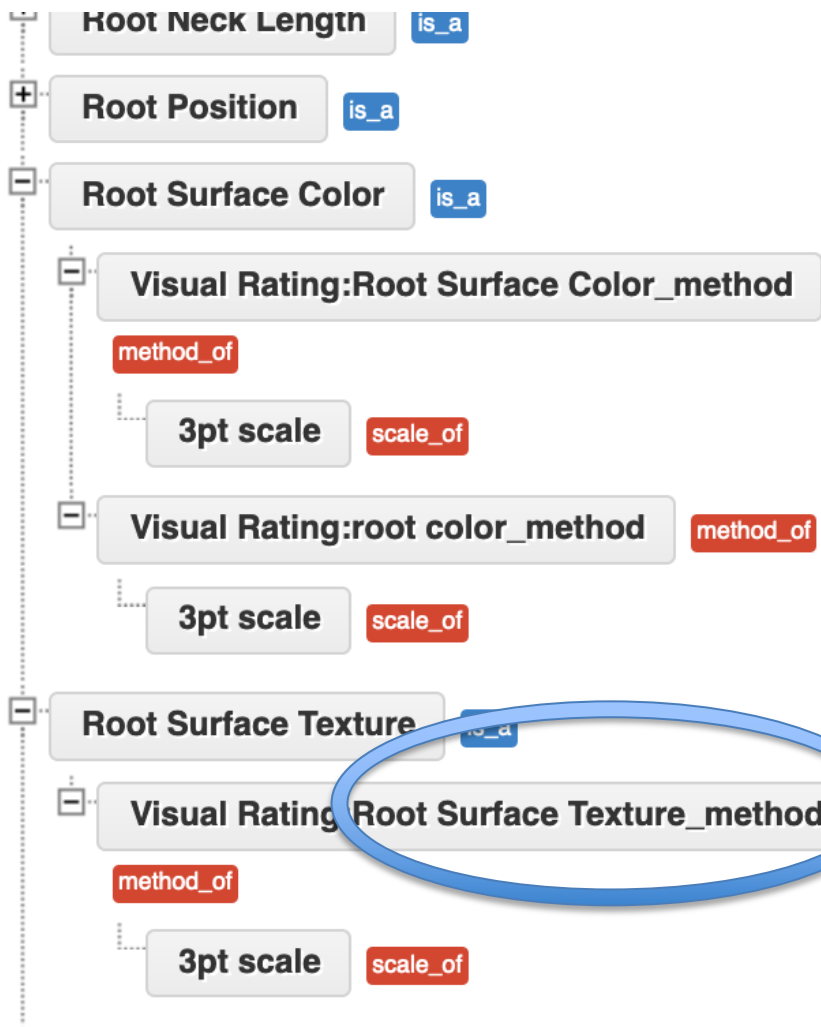


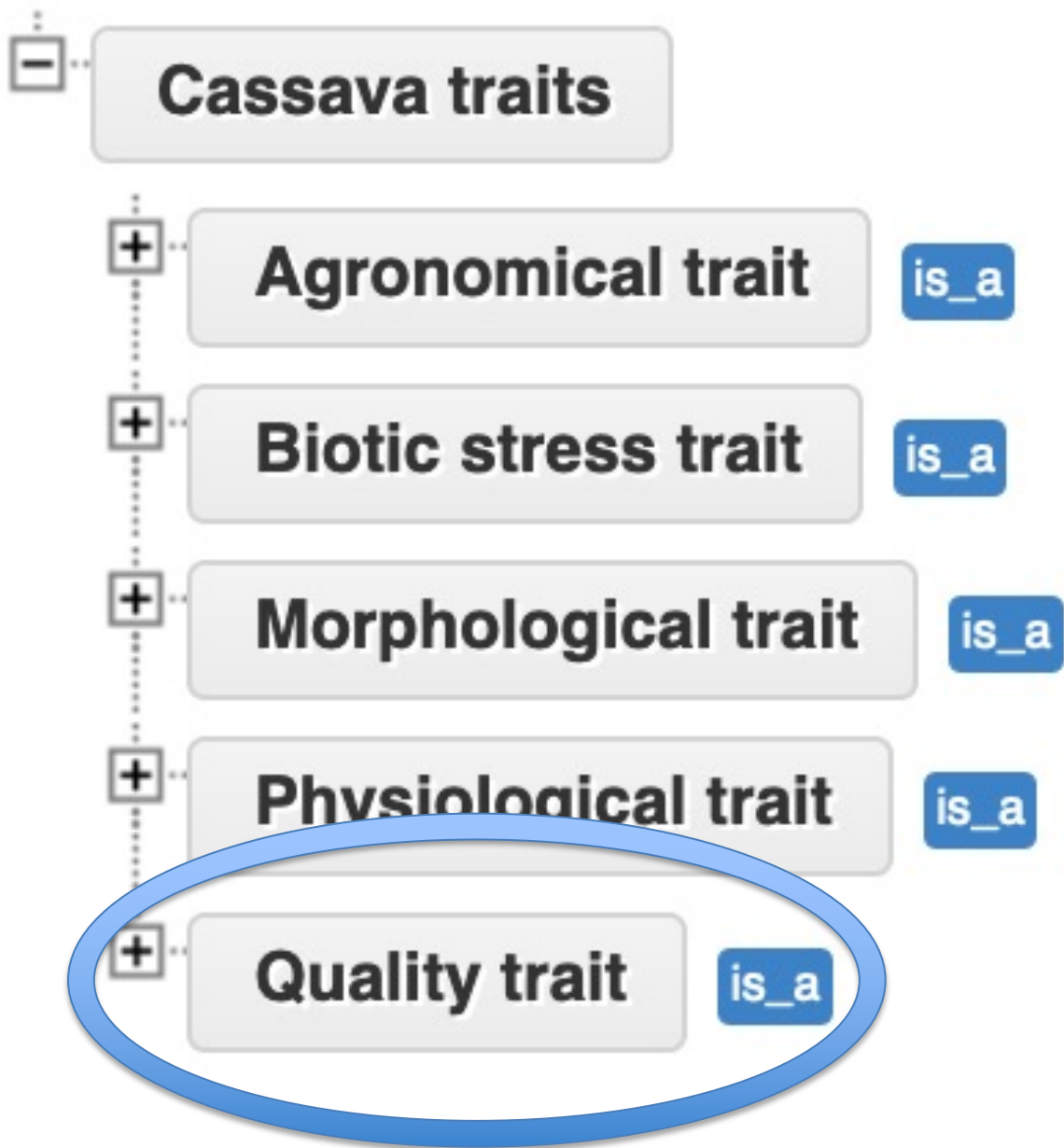


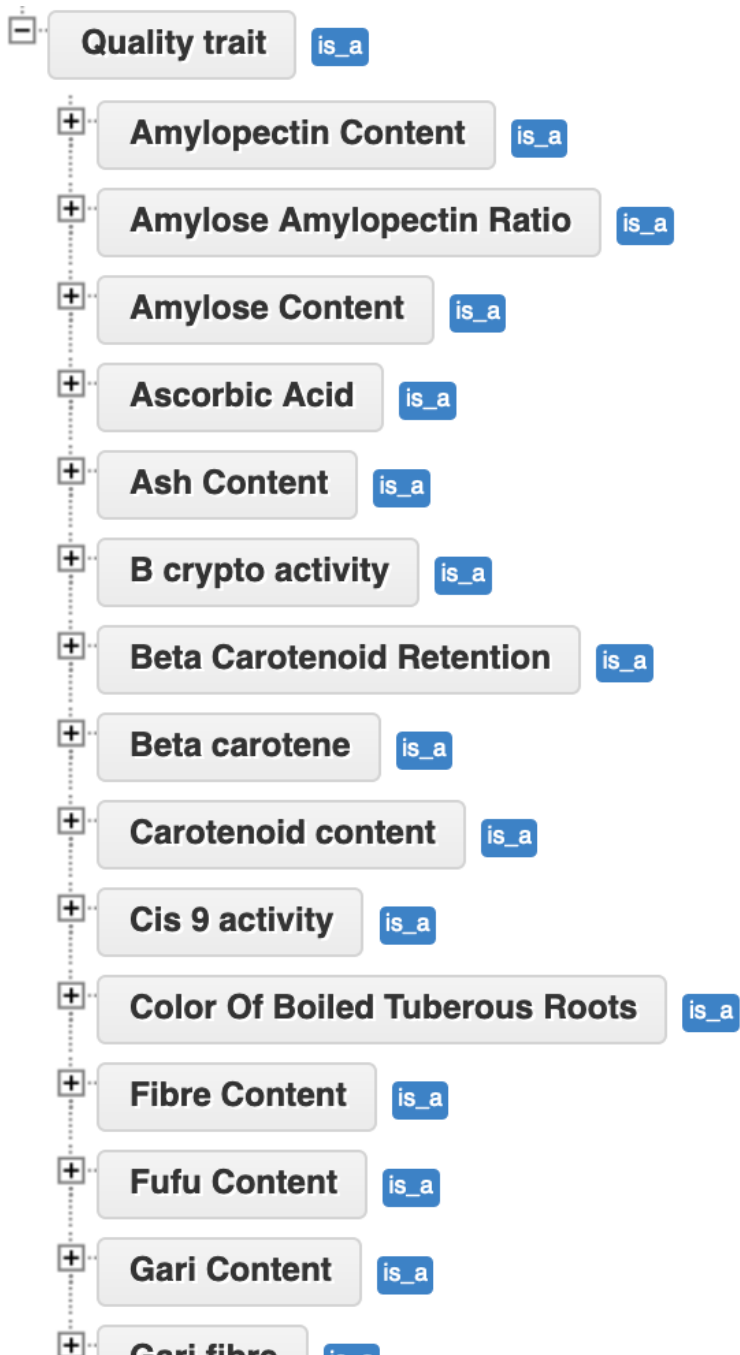


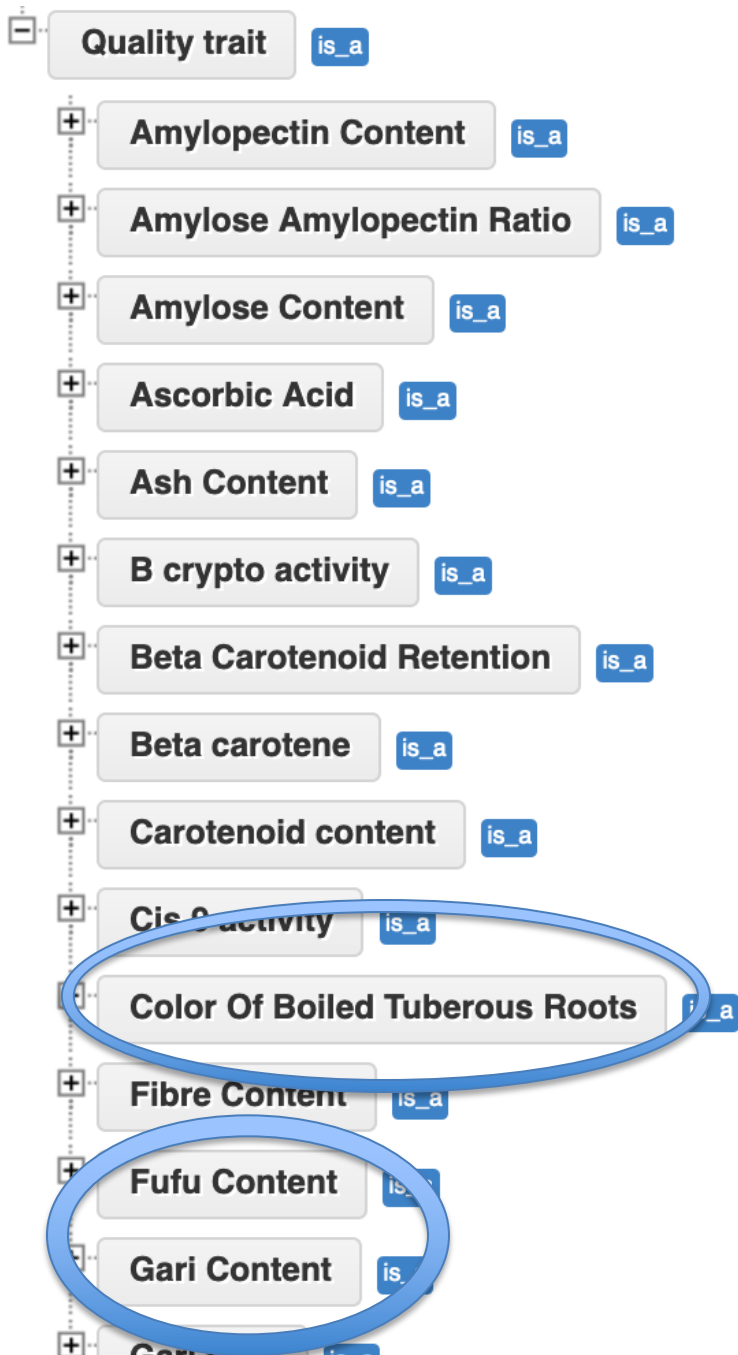


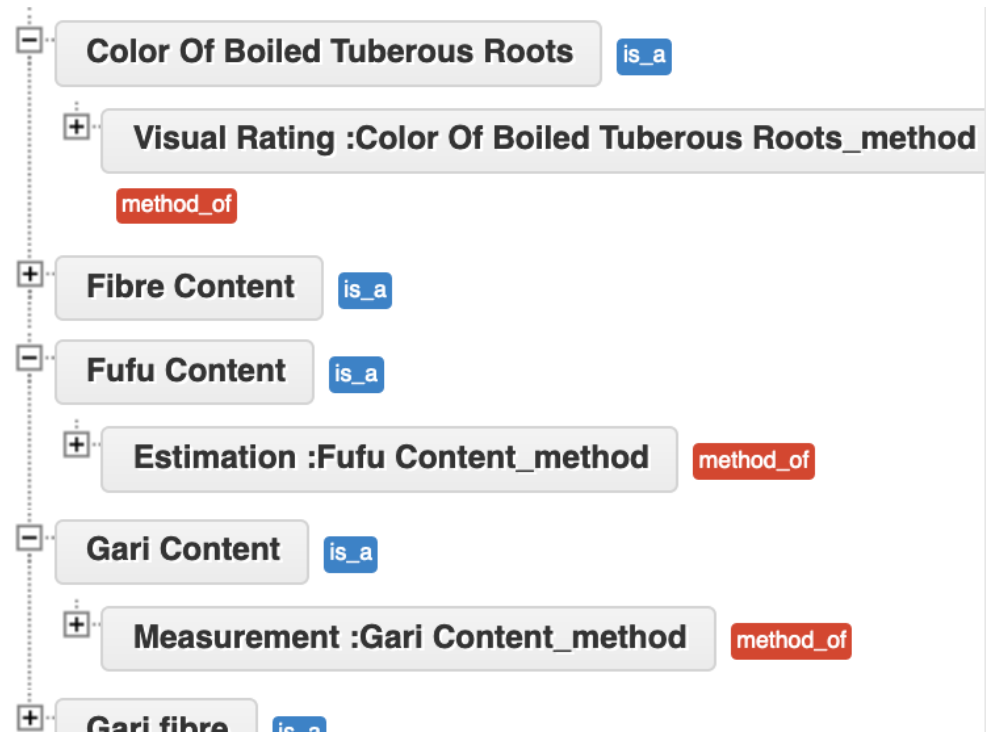
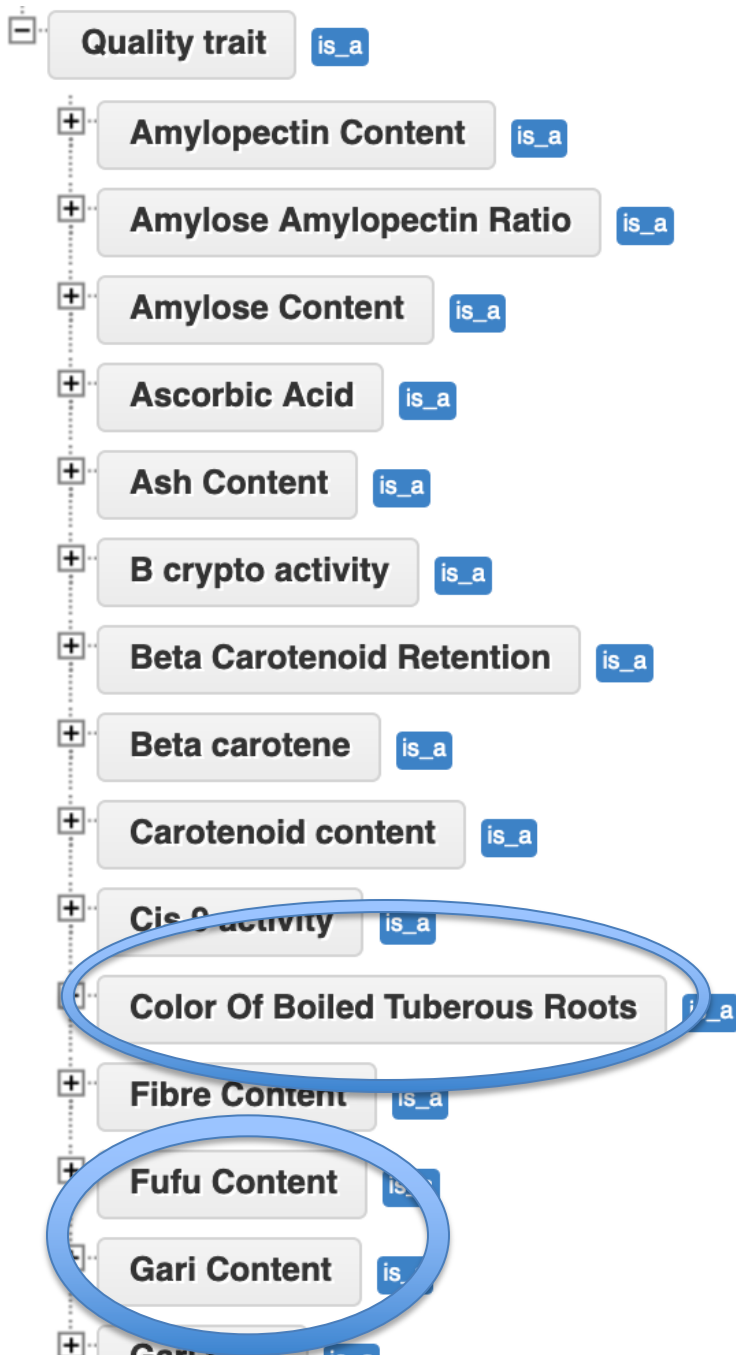










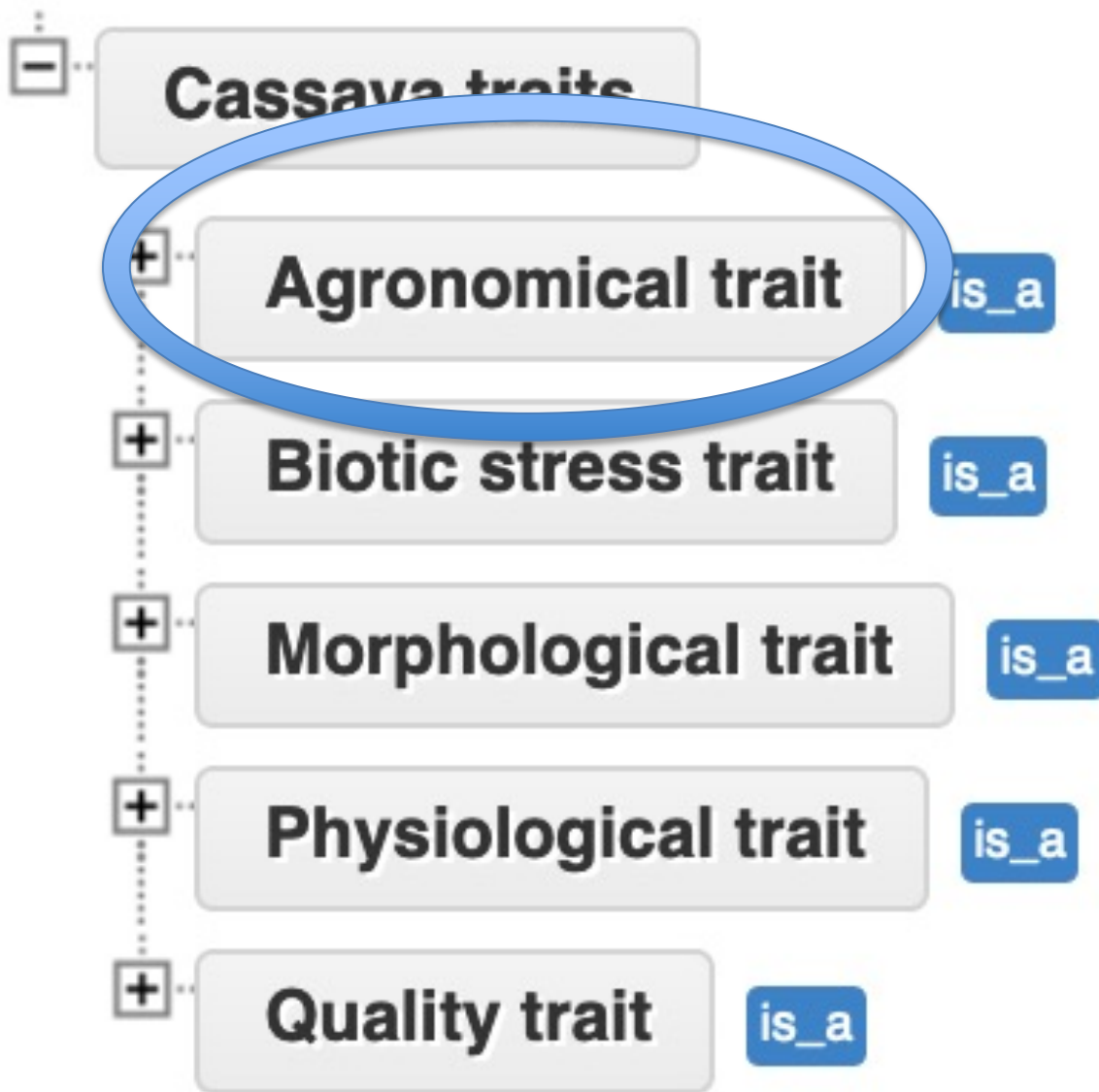


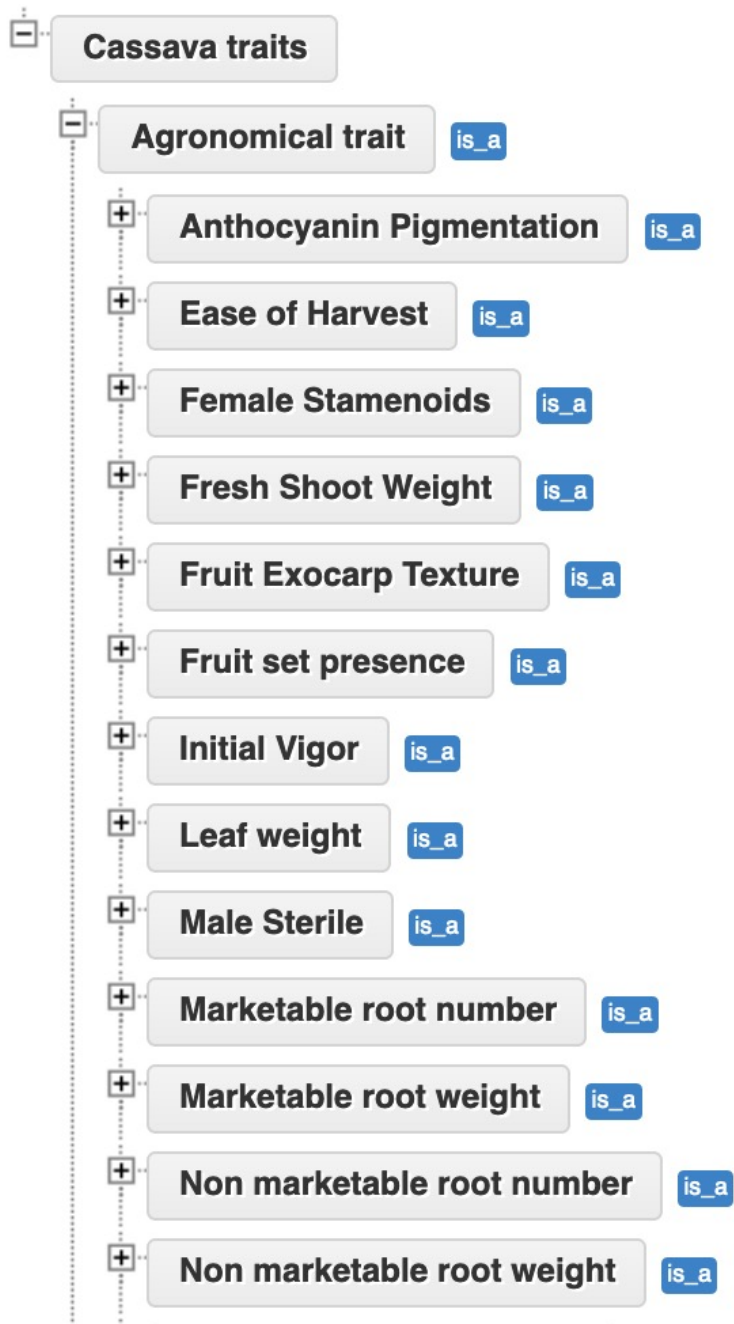
[3] Specificity vis-à-vis plants, people and history of data

Plants:

- **Each crop requires specific ontology adaptation:**
“Ontologies need to capture everything people are doing, all the methods, there is no wrong or right way”
(PI_17_A)
 - Each ontology is specific to the crop, differences are too large
 - E.g. “Color of outer surface of storage root cortex (inner skin)” for cassava
 - Organism-dependent standards, with Crop Ontology acting as a crucial “regulatory body” (R_12_B)







 Cassava traits

 Agronomical trait

 **Anthocyanin Pigmentation** 

 **Ease of Harvest** 

 **Female Stamenoids** 

 **Fresh Shoot Weight** 

 **Fruit Exocarp Texture** 

 **Fruit set presence** 

 **Initial Vigor** 

 **Leaf weight** 

 **Male Sterile** 

 **Marketable root number** 

 **Marketable root weight** 

 **Non marketable root number** 

 **Non marketable root weight** 

Marketable root number is_a

Counting:marketable root number_method

method_of

Number scale_of

Marketable root weight is_a

Measurement :marketable root weight_method

method_of

kg scale_of

Non marketable root number is_a

Counting:non marketable root number_method

method_of

Number scale_of

Non marketable root weight is_a

Measurement :non marketable root weight_method

method_of

kg scale_of

[3] Specificity vis-à-vis plants, people and history of data

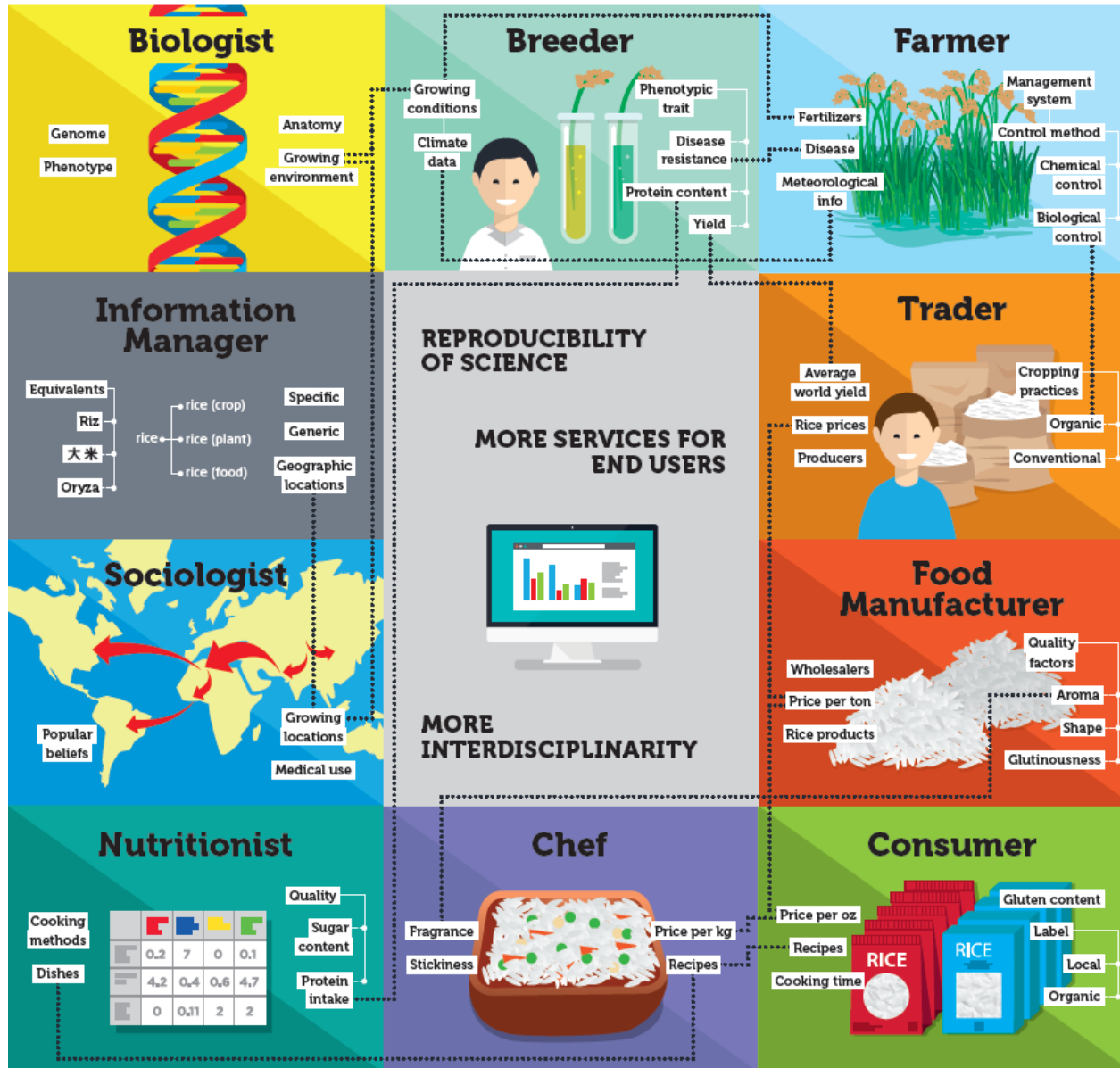
People:

- Importance of dialogue among stakeholders
 - Assessing the need to add terms through open communication with breeders
 - Yearly farm visits and dialogue over preferred traits; Cassava breeders meetings and training sessions. E.g. what is “shoot weight”?
 - Use of English vs own-language descriptions → translation as significant in shaping variation of descriptions from user to user
 - National and international coordination:
 - Cross validation of terms to avoid duplication
 - E.g. integrating cassava ontology into planteome.org project through the Crop Ontology; CGIAR Big Data Platform; Agricultural Communities of Practice

SEMANTICS - THE WAY TO RECONCILE POINTS OF VIEW AND DATA

THE EXAMPLE OF "RICE"

Research Data
Alliance(RDA)
Agrisemantics
Working Group,
2017



[3] Specificity vis-à-vis plants, people and history of data

History of Data:

- Work is “never-ending” as research continues, and plants themselves evolve and change
 - **“At no point can the ontology be static”**: always traits to add, things to adjust
- Acknowledgment of multiplicity of goals & methods in data generation and use
 - Biological understanding, conservation, trade, nutrition/food security, computational capabilities, environmental sustainability, etc...
- The perfect tool? Tracking history as more significant than standards:
 - “If you get an accession, you should trace its history, get its attributes, in which trials it has been used and its performance in the trials at every level. Quality, agrobiotics, stresses. All information should be linked to accession identifier” (crop database curator)

Situated Ontologies: Process-Sensitive Naming

Towards process-sensitive (trait) naming

- Environment takes center stage: from focus on trait in itself to *relation between trait and context*
- Naming processes not solely as application of nomenclature, but as explicitly tied to methods of data collection *and* processing
- Facilitates finer-grained look at *intra-species* diversity
- Metadata focus on documenting difference as well as similarity and problematizing typicality of traits vis-à-vis species concepts
- Spurs efforts to document history of data processing

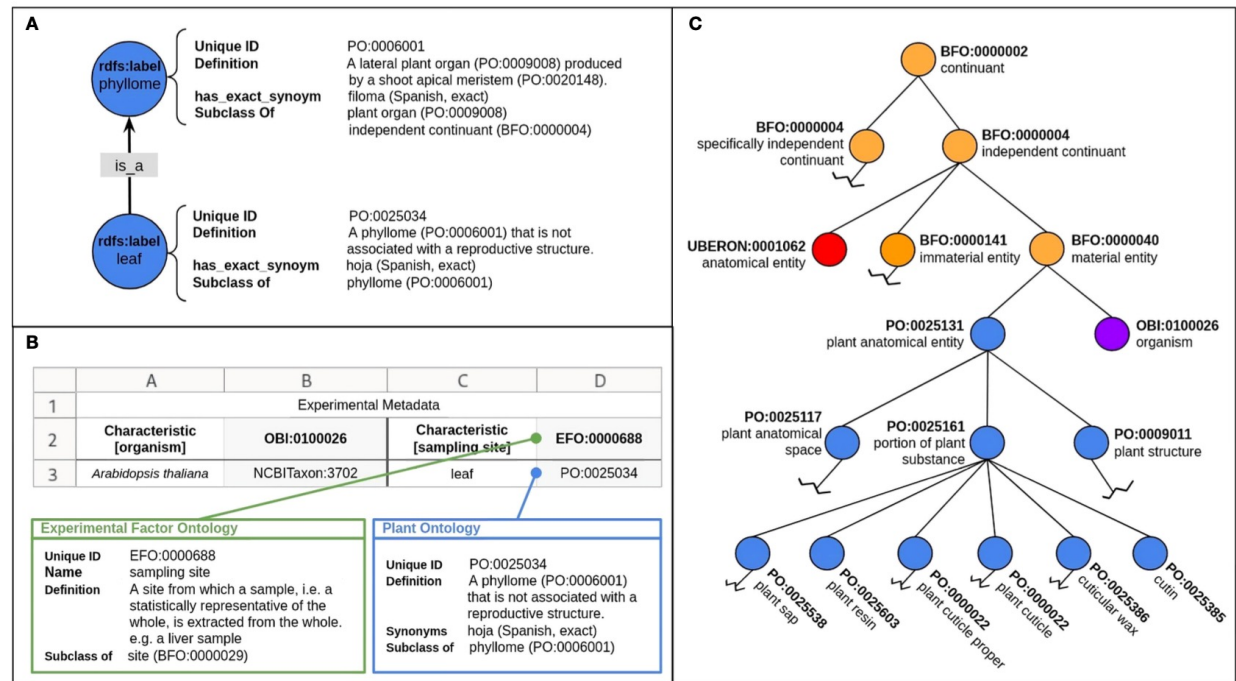
From capturing biological products to capturing the processes of cultivation and consumption

- Capture human interactions with plants and their environment, rather than plants by themselves
- Importance of the senses in identification of phenotypic traits
 - Tension with uses of language to capture and classify observations
 - Qualitative and quantitative data (e.g. tactile information on cassava pulp; colour of fruit in cassava; weight and number of bananas in banana bunch)
 - Importance of these sources for classification aimed at crops science, tension between these data types (and related germplasm collection and classifications) and formal taxonomies (and genomic-based analysis)
- Attention to the relation between valuing practices, data practices and naming practices – key to capturing ‘process’
- Arguably momentous shift: **attention to processes of data “design”, production, collection and management as key to plant data interoperability**

Ontologies for increasing the FAIRness of plant research data

Kathryn Dumschott^{1*}, Hannah Dörpholz¹,
Marie-Angélique Laporte², Dominik Brilhaus³,
Andrea Schrader⁴, Björn Usadel^{1,5}, Steffen Neumann^{6,7},
Elizabeth Arnaud² and Angela Kranz^{1*}

Continuing co-
evolution of data
curation:
A serious multi-
level challenge..

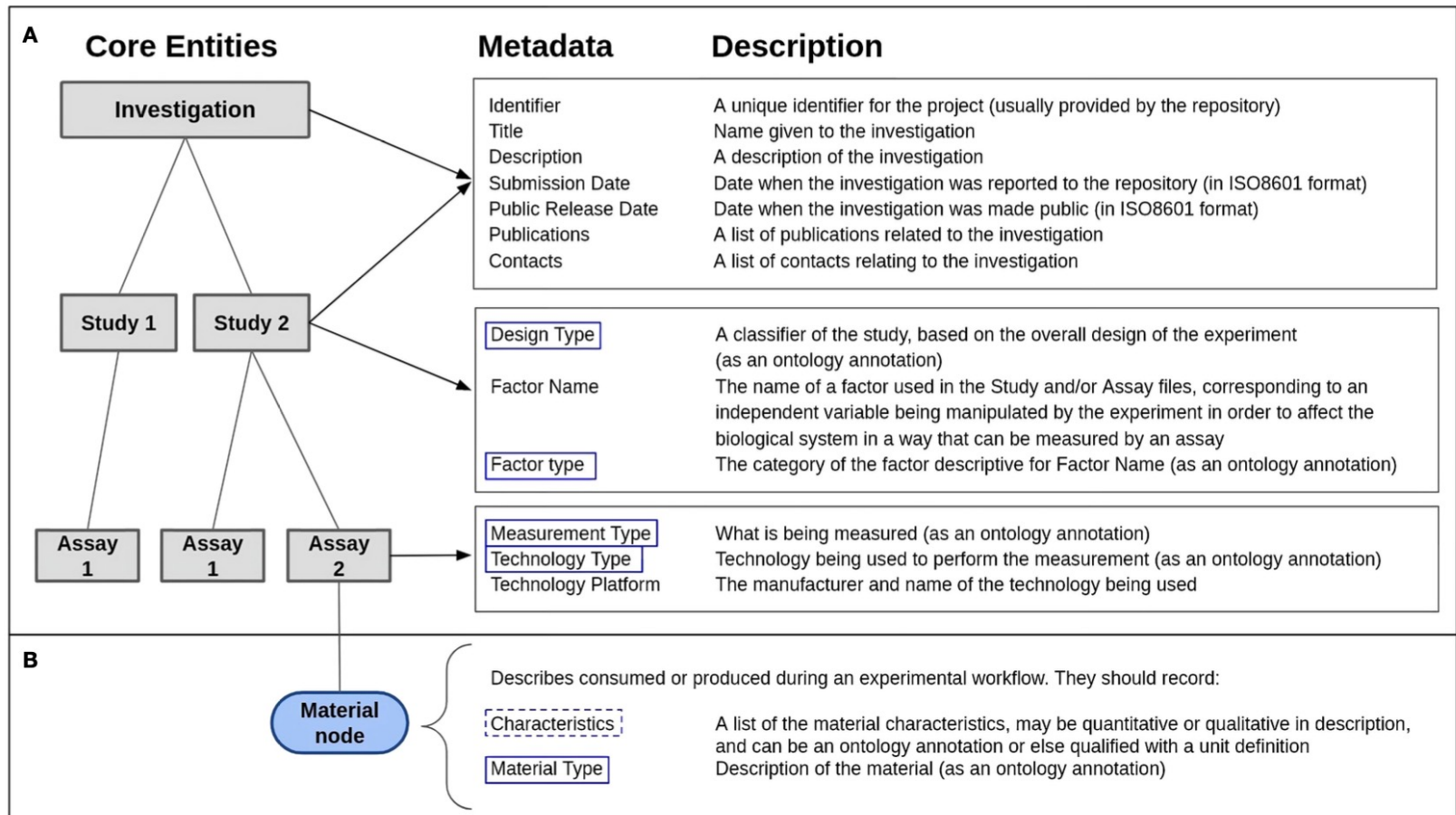


Open Biological and Biomedical Ontology Foundry

Community development of interoperable ontologies for the biological sciences

ISA - Investigation, Study, Assay Framework

.. Made worse by
intersecting
metadata
frameworks



Crop Ontology as Transnational Data Broker

- **develop** participatory practices for crop data collection and management
 - involving local breeders and farmers in developing data infrastructures, addressing gender inequality in protocols and training, and resisting privatization of resulting data through commitment to Open principles
- **mediate** between crop-specific, local databases and international initiatives in plant data management
 - many led by US/EU researchers or international agencies such as FAO
 - challenge of integrating better ontology standards (BFO etc)
- **negotiate** the tensions arising from attempts to link locally acquired digital information into global networks, and the related effort to regulate the transfer of plant genetic materials, such as germplasm, across national borders

Crop Ontology as Transnational Data Broker

Huge tensions remain around data sovereignty, extractive forms of data re-purposing & (lack of) investment in data curation

- **Benefit-sharing problems:**
 - Market control and IP regimes by agrotech
 - Debate around who should reap benefits continues with no resolution
 - Lack of trust = no sharing, no re-use
 - No participation in governance and design of tech, esp. as exponential data re-use and loss of provenance
- **Regime of agricultural development erases cultural, biological, scientific and semantic diversity. E.g.**
Molecular bias endures:
 - Molecular data rated as more reliable than observational data, and used to structure databases and data analysis – greatly limiting integration across scales / perspectives
 - Contributors with no related expertise and instruments viewed as second class
 - Legal frameworks preoccupied chiefly with Digital Sequence Information, unclear frame for other data types (sometimes thought of as meta-data)

Crop Ontology as Transnational Data Broker

Huge tensions remain around data sovereignty, extractive forms of data re-purposing & (lack of) investment in data curation

- Standards misalign with (some) goals and available expertise: Focus on elite breeds risks to exclude farmer knowledge and local diversity, as well as sustainable agriculture and agrodiversity
- Digital divide expands: uneven capacity to access and use digital infrastructure, and participate in its development and governance
- Surveillance concerns: unregulated mass surveillance of farming behaviors and local environmental characteristics
- Lack of investment in data curation in the age of ‘magical’ genAI

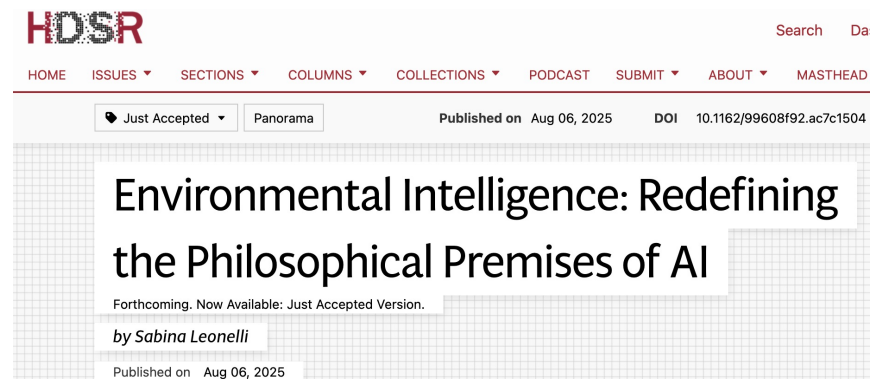
Situated, process-sensitive ontologies to support research across locations and agendas

- Crucial to forge data systems, chains of evidence, AI models and conceptual tools that
 - Bridge between different, yet relevant epistemic spaces
 - Take account of local knowledge and uses of the materials in question
 - Integrate heterogeneous data for robust and responsible geospatial reasoning
- e.g. to re-imagine agriculture away from high-yield monocultures
 - Agrobiodiversity can boost long-term sustainability of food production systems, esp. in the face of climate change
 - Conservation of forgotten/indigenous crops as key to food security
 - Plant science needs to reposition itself accordingly, starting from semantics!

Reframing the AI debate: Environmental Intelligence

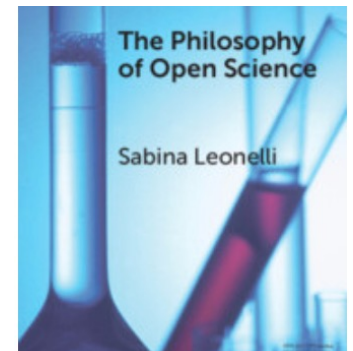
the effort to develop digital technology and data-intensive algorithmic systems to sustain and enhance life on this planet

- in full awareness of interdependence between technology and the natural world and the severe limits in human understandings of such interdependence..
- .. and thus of possible effects of technological change on the planet as a whole, particularly under conditions of climate emergency and growing social inequities
- “EI can respond to such conditions by prioritizing innovation that bolsters biological, social and scientific diversity, and **explicitly designing interventions to enhance agency and representation of the most vulnerable in human societies and to provide safeguards for non-human forms of life**” (Leonelli 2025)



Reframing the openness debate: Humane Open Science

- Discovery as skilled, distributed interaction with the world
 - centrality of **connections** (intellectual, material, emotional) among research participants as backbone to communication, constructive critique and creative exchange
 - focus on **social agency**
 - creating new intimacies, potentially facilitating trust and collaboration
 - **justice** and **diversity** as crucial conditions for inquiry
- Connections need to be *judicious*:
 - Situated and responsive to context
 - **Determining what constitutes relevant context is key part of any investigation**

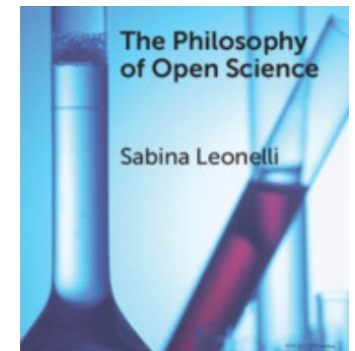


Reframing the openness debate: Humane Open Science

- about **responsible use**
- about the critical and constructive scrutiny of how **digital platforms** can support existing and future work
- **good for some and not others**: value-judgements and choices are unavoidable when developing open research and infrastructures
- **accessible to some and not others**: transparent criteria for which users are privileged can be a platform for trustworthiness
- facilitating **equity** in research production and consumption



Critical role of situated, process-sensitive ontologies

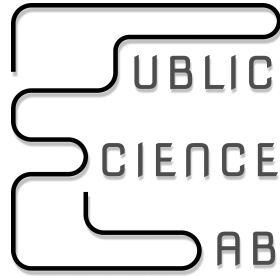


Thank you for your attention

TransforM

Munich Center for
Transformative Technologies
and Societal Change

From more innovation to better innovation.



European Research Council

Established by the European Commission



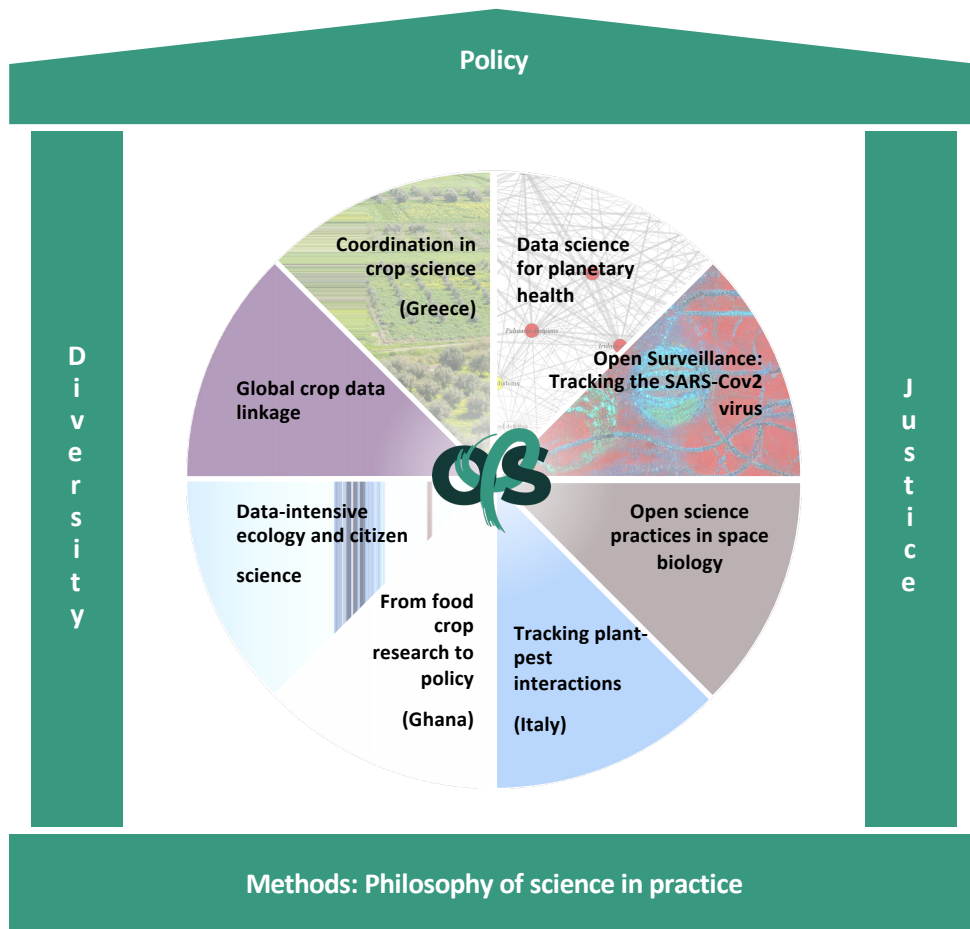
FOR2026-MUNICH CONFERENCE 4-6 MAY 2026

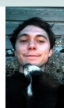
The Future of Open Research: Reliable, Responsible, Equitable

Ethical Data Initiative Virtual Townhall

November 28 @ 1:30 pm – 4:30 pm





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abstract

Are ontologies a tool to remove or to highlight the role of human judgement in data systems? This is the question posed in this lecture, which reviews efforts to build ontologies as tools to automate data analysis – or at least make it amenable to computational methods – and argues that ontologies can help to identify and articulate the role and significance of human intervention, values and goals in making sense of data. I exemplify this argument through a study of the Crop Ontology and its application to the study of plant traits. This ontology exemplifies what I call a process-sensitive approach to data semantics, in which the naming of the phenomena in question (plant traits) attempts to capture the environmental interactions (including those between plants, researchers and wider publics) instead of the biological products in and of themselves. Reference: Leonelli, S. (2022) Process-Sensitive Naming: Trait Descriptors and the Shifting Semantics of Plant (Data) Science.

Philosophy, Theory and Practice in Biology 14:16

<https://doi.org/10.3998/ptpbio.16039257.000000>

Data absences and data abundance

- Huge asymmetries!

Datafication 2.0

the challenge of interpreting absence

“What is absent matters, and yet our current understanding of the physical universe suggests that it should not. A causal role for absence seems to be absent from the natural sciences” (Terrence Deacon 2005, 3)

Looking beyond the given: Data shadows

- **Absent data** - not present or readily available
 - Gaps in the record; lost, overlooked or unusable data
- **Imagined data** - willed into being as projections of epistemic expectations, whether or not those fit research conditions
 - Synthetic data, simulations
- **'Bad' data** - inadequate to the purposes and conventions for which they are being considered as potential evidence
 - Anecdotal medical observations
- **Negative data** - failing to corroborate a given hypothesis or add new information to a given line of inquiry
 - drugs failing effectiveness tests
 - failure to replicate

From Convenient AI to Environmental Intelligence

From Convenient AI to Environmental Intelligence

Reframing the AI debate: **Environmental Intelligence**

the effort to develop digital technology and data-intensive algorithmic systems to sustain and enhance life on this planet

- in full awareness of interdependence between technology and the natural world and the severe limits in human understandings of such interdependence..
- .. and thus of possible effects of technological change on the planet as a whole, particularly under conditions of climate emergency and growing social inequities
- “EI can respond to such conditions by prioritizing innovation that bolsters biological, social and scientific diversity, and explicitly designing interventions to enhance agency and representation of the most vulnerable in human societies and to provide safeguards for non-human forms of life”

Leonelli (2025) Environmental Intelligence: Redefining the Philosophical Premises of AI. *Harvard Data Science Review*.

“Convenience AI” versus Environmental Intelligence

1. *Intelligence is distributed, embodied and situated*
2. *Discriminating what counts as relevant information for decision-making is a human responsibility*
3. *Attempts to control the natural environment have self-destructive effects*
4. *It is imperative to address the climate crisis and related existential threats to human and nonhuman life, including the unsustainable development of AI*
5. *Social and economic inequities are highly entrenched and have deleterious environmental effects*
6. *Preserving and fostering diversity is core to human experience and life on the planet*

1. Intelligence is generalizable and disembodied

2. AI is grounded on robust evidence base and can discriminate what is relevant to which problem

3. AI needs to achieve control of the environment

Thank you for your attention



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